Deterministic Distance-Based Source Sampling for Efficient Multi-Scale Network Centrality Analysis

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Abstract

Computing localised network centrality (closeness and betweenness) at multiple distance scales becomes prohibitively expensive at metropolitan scales. Source sampling reduces this cost, but comparative urban analysis requires a deterministic, network-agnostic schedule so that rankings remain directly comparable across locations.

We derive such a schedule $p(d)$ from the Hoeffding bound applied to a canonical grid reach model. One user-chosen parameter ε (default 0.06) controls the accuracy–speed trade-off; the same schedule serves both metrics. Validation on the Greater London (156,285 nodes) and Greater Madrid street networks at 1–20 km confirms rank preservation ($\rho \geq 0.97$) with speedups of up to 8.0×. The approach is implemented in the open-source `cityseer` Python package.

Keywords: network centrality, sampling, urban morphology, betweenness, harmonic closeness, computational efficiency

1 Introduction

Urban morphological analysis relies on localised network centrality to reveal patterns of accessibility and flow in street networks. Closeness and betweenness are computed within distance thresholds ranging from pedestrian (∼500 m) to metropolitan (∼20 km) scales, with computational cost driven by the longest distances. For a network of 156,285 nodes, a multi-scale analysis spanning ten distance thresholds takes several hours.

Source sampling exploits the fact that each source traversal contributes to the centrality estimates of all nodes it reaches: running traversals from a random subset and rescaling yields approximate estimates. The mean number of reachable nodes (*reach*, r) governs both cost and accuracy: at short distances reach is small and traversals are cheap; at long distances reach is large and each estimate draws on many source contributions, so sampling yields the largest speedups precisely where accuracy is easiest to maintain.

Comparative urban analysis requires the sampling schedule to be *deterministic* and *network-agnostic*: the same analysis distance must produce the same sampling probability regardless of network. Adaptive methods such as KADABRA (Borassi and Natale, 2019) provide convergence guarantees but produce network-dependent sample counts, precluding direct cross-network comparison.

For global closeness, the Eppstein–Wang source-sampling bound (Eppstein and Wang, 2004) motivates replacing node count n with mean reach r, giving $k = O(\log r/\varepsilon^2)$. Path-sampling methods (Riondato and Kornaropoulos, 2016) work well for global betweenness but lose their cost advantage under localisation, where a single Brandes traversal (Brandes, 2001) updates all nodes in the bounded neighbourhood (Section 3). We therefore use a single deterministic source-sampling schedule $p(d)$ for both metrics.

Our contributions are:

1. A deterministic distance-only source-sampling schedule $p(d)$, derived from the Hoeffding bound applied to a canonical grid reach model, serving both closeness and betweenness via shared Brandes traversals.

2. Empirical validation on the Greater London (156,285 nodes) and Greater Madrid networks at 1–20 km with $\varepsilon = 0.06$, demonstrating rank preservation ($\rho \geq 0.97$) with speedups of up to 8.0×.

The approach is implemented in the open-source `cityseer` Python package.

2 Methods

We consider two centrality measures, both computed with a distance cutoff $d_{\max}$ as is common in urban network analysis.

Harmonic closeness (Rochat, 2009; Freeman, 1979) sums inverse distances to all nodes within the threshold:

$$C_H(v;\, d_{\max}) = \sum_{\substack{u \neq v \\ d(v,u) \leq d_{\max}}} \frac{1}{d(v,u)} \tag{1}$$

Betweenness (Freeman, 1977) counts the fraction of shortest paths (with total length at most $d_{\max}$) passing through each node:

$$C_B(v;\, d_{\max}) = \sum_{\substack{s \neq t,\, s \neq v,\, t \neq v \\ d(s,t) \leq d_{\max}}} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

where σ_{st} is the number of shortest paths from s to t, and $\sigma_{st}(v)$ is the number passing through v.

Reach r is the mean number of nodes within $d_{\max}$:

$$r = \frac{1}{N} \sum_{v} |\{u : d(v,u) \leq d_{\max}\}| \tag{3}$$

In a canonical grid network with inter-node spacing $s = 175$ m, reach grows from approximately 100 nodes at 1 km to over 40,000 at 20 km.

Synthetic networks. We generate synthetic networks using 3 topology templates: trellis (high connectivity, representing planned urban cores), tree (low connectivity, representing suburban sprawl), and linear (intermediate connectivity). These represent a range of urban street network structures. Each topology is tiled to create networks of approximately 12 km extent; we evaluate accuracy on a central 4 km×4 km window to reduce edge effects. We test 4 distance thresholds from 250 m to 4,000 m, and 3 error tolerances (ε) from 0.01 to 0.5.

Boundary effects. Nodes near network edges have artificially lower centrality because their catchments are truncated. We apply an inward buffer from the convex hull equal to the maximum analysis distance; only nodes inside this buffer are included in accuracy calculations. This is consistent with common practice for localised network analysis (Gil, 2017).

Spearman rank correlation (ρ) between sampled and ground-truth centrality values:

$$\rho = 1 - \frac{6 \sum_i (R_i^{\text{sampled}} - R_i^{\text{true}})^2}{N(N^2 - 1)} \tag{4}$$

where R_i denotes the rank of node i. Spearman correlation is appropriate because urban applications primarily use *relative* rankings (e.g., identifying the most central streets) rather than absolute values.

Normalised additive error $\varepsilon_{\text{obs}} = \max_v |C^{\text{sampled}}(v) - C^{\text{true}}(v)| \,/\, C_{\max}$, where $C_{\max}$ is an upper bound on the centrality value within the distance-bounded neighbourhood: $r(r-1)$ for betweenness and r for harmonic closeness. This normalised error lies in $[0, 1]$ and corresponds to the ε parameter used to set the sampling schedule.

Validation networks. We validate on two real-world street networks: the Greater London Authority (GLA) road network (Ordnance Survey Open Roads; $\approx$156,285 nodes after preprocessing) and the Greater Madrid metropolitan network. Both use a 20 km inward buffer to eliminate edge artifacts.

3 A Unified Sampling Model and Deterministic Distance Schedule

On synthetic networks, Spearman ρ exceeds 0.95 well before ε reaches the default of 0.06 for both closeness and betweenness (Figure 1).

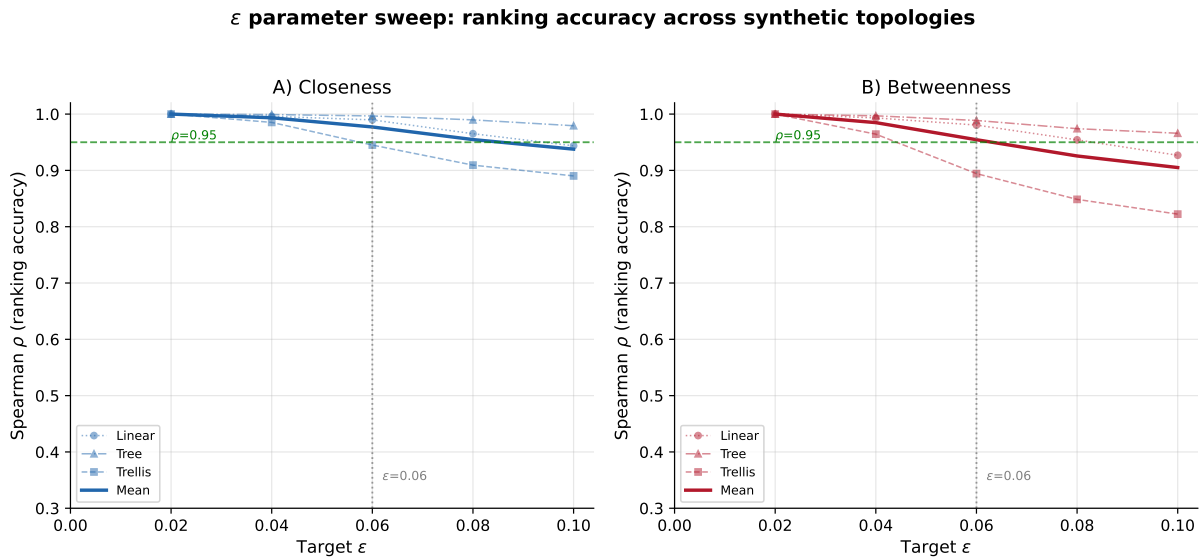

Figure 1: **The sampling opportunity.** Ranking accuracy (Spearman ρ) vs. error tolerance ε for closeness (A) and betweenness (B), averaged across synthetic topologies and distances. The green dashed line marks $\rho = 0.95$; the grey dotted line marks the paper default $\varepsilon = 0.06$. Shaded bands show 95% confidence intervals.

Closeness: Hoeffding source-sampling. Following Eppstein and Wang (Eppstein and Wang, 2004), each of k uniformly sampled sources contributes a normalised value in $[0, 1]$ to each target node's closeness estimate. The Hoeffding bound (Hoeffding, 1963) with a union bound over target nodes yields sample complexity $k = \log(2n/\delta)/(2\varepsilon^2)$ for global centrality. Under localisation with threshold $d_{\max}$, replacing n with mean reach r gives

$$k = \frac{\log(2r/\delta)}{2\varepsilon^2} \tag{5}$$

with corresponding sampling probability

$$p = \min\left(1,\ \frac{k}{r}\right) \tag{6}$$

This model has no fitted parameters: ε (error tolerance) and δ (failure probability) are user-chosen conventions. At default values $\varepsilon = 0.06$, $\delta = 0.1$:

- **Low reach** ($r \lesssim 1{,}600$): $k > r$, so $p = 1$ and computation is exact (the schedule saturates naturally).
- **High reach**: p decreases as $O(\log r/r)$, so speedup $1/p$ grows as $O(r/\log r)$.

From reach-based to deterministic sampling probability. Equations 5–6 express p as a function of reach r, which varies across networks. To eliminate this dependence, we substitute a canonical grid model: $r \approx \pi d^2/s^2$ with spacing $s = 175\,\mathrm{m}$ (the upper end of typical European block lengths, representing sparse morphology). Since most urban networks are denser, the canonical reach underestimates actual reach, so the prescribed p is conservative. This is the schedule implemented in `cityseer`.

Betweenness: path-sampling under localisation. Betweenness contributions are sparser and heavier-tailed than closeness contributions. Path-sampling methods (Riondato and Kornaropoulos, 2016) exploit this on *global* graphs by sampling individual shortest paths, with

$$m = O\left(\frac{\log \mathrm{VD} + \log(1/\delta)}{\varepsilon^2}\right) \tag{7}$$

sampled paths. Under a localisation cutoff $d_{\max}$, however, the admissible pair set depends on local morphology (no fixed global pair space), and each path sample updates betweenness along only one path, while one source traversal with Brandes backpropagation updates all r nodes in the bounded neighbourhood.

A cost model makes the crossover explicit. Let $m(d)$ be the path-sample budget, $c_{\mathrm{pair}}(d)$ the cost per path sample, and $c_{\mathrm{src}}(d)$ the cost per bounded source traversal. Then

$$T_{\mathrm{path}}(d) \approx m(d)\, c_{\mathrm{pair}}(d), \qquad T_{\mathrm{exact}}(d) \approx n\, c_{\mathrm{src}}(d). \tag{8}$$

Path-sampling is slower than exact computation when

$$m(d) \geq m^\star(d) := n\, \frac{c_{\mathrm{src}}(d)}{c_{\mathrm{pair}}(d)}. \tag{9}$$

Since both cost terms depend on morphology, $m^\star(d)$ is not a portable function of distance alone, precluding a robust switching rule.

Source-sampling resolves betweenness without additional sampling cost. A single bounded Dijkstra traversal from source s with Brandes backpropagation accumulates dependency contributions[1] $\delta_s(v)$ for all r reached nodes. Each $\delta_s(v)$ is bounded in $[0, r-1]$, and maximum betweenness within the neighbourhood is at most $r(r-1)$. After normalising by $C_{\max}$, the per-source contribution lies in $[0, 1/r]$, and the same Hoeffding argument yields sample complexity identical to the closeness case (Equations 5–6). The bound is conservative for two reasons: dependency values across target nodes are not independent (sharing a shortest-path tree), and betweenness contributions are more variable than closeness contributions, making the range-based bound less tight. This explains the empirically thinner margins for betweenness (Section 4).

Table 1 provides the deterministic sampling probabilities at standard analysis distances for several ε values.

Error structure across reachability. Figure 2 shows a crossover: absolute error increases with reach, but normalised error decreases. High-reach nodes have higher effective sample sizes, so their relative errors are small. We term this *precision scales with importance*: the nodes most relevant to urban analysis receive the highest relative precision.

Table 1: Deterministic sampling schedule at standard analysis distances under different error tolerances ($\delta = 0.1$, $s = 175\,\mathrm{m}$). Paper default: $\varepsilon = 0.06$ for both metrics.

Distance	$\varepsilon = 0.02$		$\varepsilon = 0.04$		$\varepsilon = 0.06$		$\varepsilon = 0.08$		$\varepsilon = 0.1$	
	k	p	k	p	k	p	k	p	k	p
500 m	exact		exact		exact		exact		exact	
1 km	exact		exact		exact		exact		exact	
2 km	exact		exact		exact		exact		exact	
5 km	exact		exact		1,506	58.7%	847	33.0%	542	21.1%
10 km	exact		3,822	37.3%	1,698	16.6%	955	9.3%	611	6.0%
20 km	17,022	41.5%	4,255	10.4%	1,891	4.6%	1,063	2.6%	680	1.7%

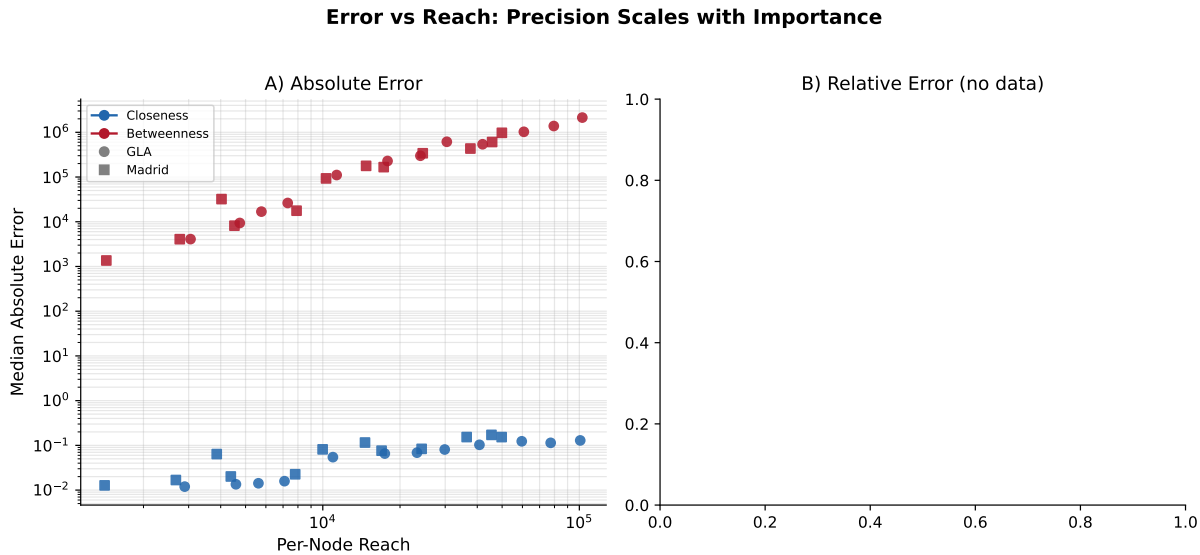

Figure 2: **Error crossover: precision scales with importance.** Per-node median errors pooled across reach quartiles at distances where $p < 1$, for Greater London (circles) and Greater Madrid (squares). (A) Absolute error increases with reach. (B) Normalised error (divided by r for closeness, $r(r-1)$ for betweenness) decreases with reach.

Table 2: Sampling approaches for centrality approximation. All prior methods target global centrality; our approach replaces node count n with distance-bounded reach r and applies one schedule to both metrics.

Method	**Metric**	**Guarantee**	**Sample complexity**	**Key parameter**
Eppstein & Wang (2004)	Closeness	Additive, all nodes	$O(\log n \,/\, \varepsilon^2)$	Node count n
Bader et al. (2007)	Betweenness	Relative, per vertex	$O(1 \,/\, \varepsilon^2 b(v))$	Betweenness $b(v)$
Riondato & Korn. (2016)	Betweenness	Additive, all nodes	$O((\log \mathrm{VD} + \log(1/\delta)) \,/\, \varepsilon^2)$	Vertex diam. VD
KADABRA (2019)	Betweenness	Additive, all nodes	Adaptive	Diameter D
SILVAN (2023)	Betweenness	Non-uniform	Variance-dependent	Path variance
This work	**Both**	**Additive, all nodes**	$\boldsymbol{O(\log r \,/\, \varepsilon^2)}$	**Reach** r

Relationship to prior bounds. Table 2 summarises existing sampling methods.

The schedule at $\varepsilon = 0.06$ is empirically conservative: observed errors (Figure 2B) are well below the Hoeffding worst case. Three sources of slack account for this: (i) the Hoeffding bound uses only the range, ignoring the smaller actual variance; (ii) the union bound does not exploit positive dependence among node-level errors; (iii) Spearman ρ is more forgiving than additive error, since small absolute errors rarely alter rankings among well-separated nodes.

Algorithm 1 Deterministic Per-Distance Sampling for Network Centrality

Require: Network $G = (V, E)$, distance thresholds $D = \{d_1, \ldots, d_m\}$
Require: Error tolerance ε (default 0.06), confidence δ (default 0.1)
Require: Canonical grid spacing s (default 175 m)
Ensure: Closeness estimates $\hat{C}_H[d]$ and betweenness estimates $\hat{C}_B[d]$ for each $d \in D$

```
for each distance d ∈ D do
    Compute canonical reach: r_d ← πd²/s²
    Compute required sources: k_d ← log(2r_d/δ)/(2ε²)
    Compute sampling probability: p_d ← min(1, k_d/r_d)
    Sample sources S_d ⊆ V by including each node independently with probability p_d
    for each source s ∈ S_d do
        Run distance-bounded Dijkstra from s; accumulate closeness via inverse distances
        Run Brandes backpropagation; accumulate betweenness path counts
    end for
    Scale both estimates by 1/p_d
end for
return {Ĉ_H[d], Ĉ_B[d] : d ∈ D}
```

4 Validation

We evaluate on the GLA and Greater Madrid networks using Algorithm 1 with $\varepsilon = 0.06$ and $\delta = 0.1$, reporting the mean over three random seeds. Ground truth uses all sources ($p = 1.0$).

Greater London. Table 3 reports results at five distances from 1–20 km. Both metrics achieve $\rho > 0.95$ at all distances, with minimum $\rho = 0.9813$ (betweenness, 20 km). Wall-clock speedup reaches 8.0× at 20 km.

Table 3: Sampling validation on Greater London network ($\varepsilon = 0.06$, $\delta = 0.1$, $s = 175$ m).

Dist.	p	ρ_c	**Spd**$_c$	ρ_b	**Spd**$_b$
1 km	100.0%	1.0000	0.4×	1.0000	0.5×
2 km	100.0%	1.0000	0.5×	1.0000	0.8×
5 km	58.7%	0.9957	0.8×	0.9977	0.7×
10 km	16.6%	0.9938	3.5×	0.9893	2.6×
20 km	4.6%	0.9932	14.7×	0.9793	15.9×

Network: Greater London Area. Deterministic distance-based schedule: same p for both metrics at each distance. Subscripts: c = closeness, b = betweenness.

Betweenness is consistently the binding constraint ($\rho_{\text{betw}} < \rho_{\text{close}}$ at every distance), as expected from its heavier-tailed contribution distribution.

Greater Madrid. Madrid provides a stricter test: its metropolitan area includes sparser peripheral zones where actual reach falls below the canonical curve (Figure 5). Despite this,

[1] Brandes' notation $\delta_s(v)$; not to be confused with the failure probability δ.

the schedule achieves $\rho > 0.95$ for both metrics at all tested distances (Table 4), with minimum $\rho = 0.9767$ (betweenness, 20 km). Wall-clock speedup reaches 7.7× at 20 km.

Table 4: Sampling validation on Greater Madrid network ($\varepsilon = 0.06$, $\delta = 0.1$, $s = 175$ m).

Dist.	p	ρ_c	$\mathbf{Spd}_c$	ρ_b	$\mathbf{Spd}_b$
1 km	100.0%	1.0000	0.6×	1.0000	0.6×
2 km	100.0%	1.0000	0.6×	1.0000	0.6×
5 km	58.7%	0.9966	0.8×	0.9980	0.8×
10 km	16.6%	0.9914	3.0×	0.9893	3.2×
20 km	4.6%	0.9819	12.4×	0.9720	13.3×

Network: Greater Madrid, 64,609 nodes. Deterministic distance-based schedule: same p for both metrics at each distance.

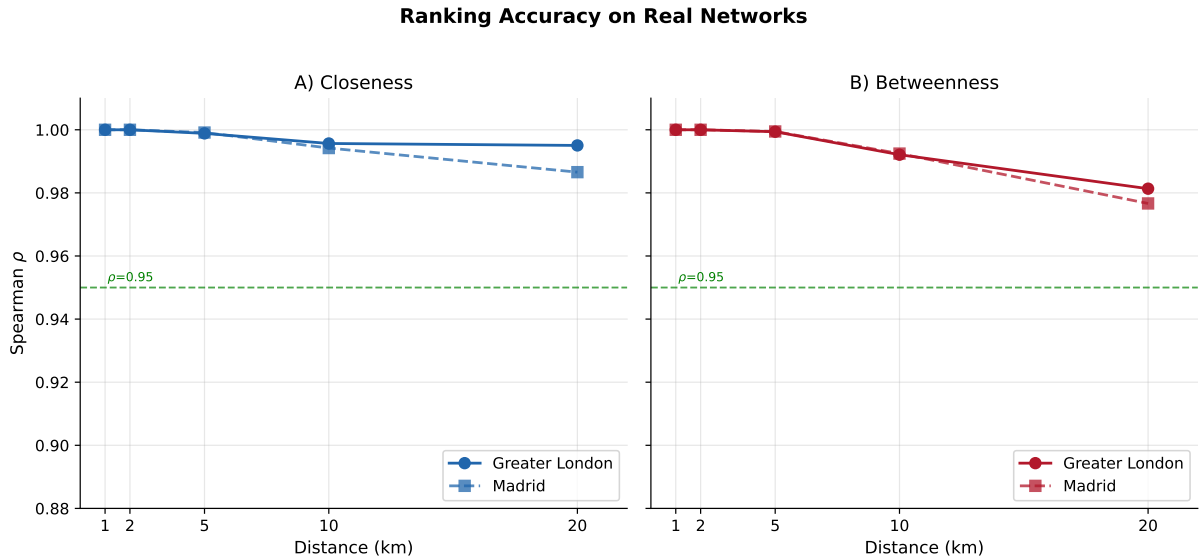

Figure 3: **Ranking accuracy on real networks.** Spearman ρ vs. analysis distance for Greater London (circles, solid) and Greater Madrid (squares, dashed), for closeness (A) and betweenness (B). The green dashed line marks $\rho = 0.95$. Both networks exceed the target at all distances; betweenness is the binding constraint.

5 Discussion and Conclusion

Practical guidance. For users of the `cityseer` library, sampling is applied automatically: the user sets ε and the library derives $p(d)$ for each distance threshold, applying it to both closeness and betweenness. For manual implementations, Algorithm 1 gives the complete procedure. Table 5 and Figure 6 provide the deterministic schedule at standard analysis distances.

At the default $\varepsilon = 0.06$:

- $\leq$4 km: $p = 1.0$ (exact; no speedup).
- 5 km: $p \approx 59\%$ (1.7× theoretical speedup).
- 10 km: $p \approx 17\%$ (6×).
- 20 km: $p \approx 5\%$ (∼8.0× measured).

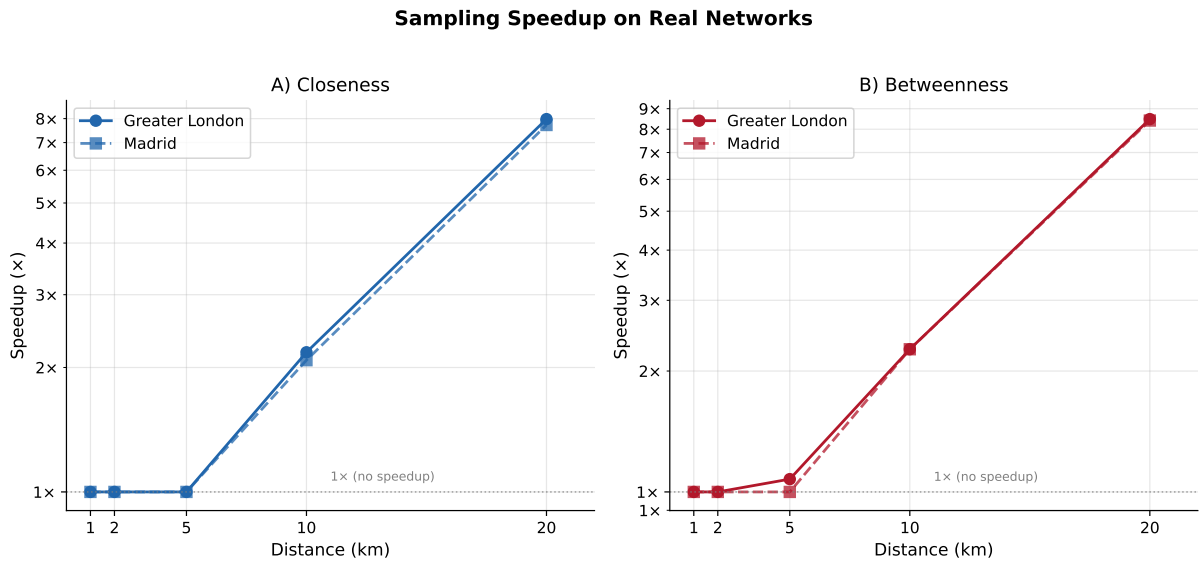

Figure 4: **Measured wall-clock speedup on real networks.** Speedup (baseline time / sampled time, log scale) vs. analysis distance for Greater London (circles, solid) and Greater Madrid (squares, dashed), for closeness (A) and betweenness (B). At short distances where p is close to 1, sampling overhead dominates and no wall-clock speedup is observed; meaningful speedup emerges at 10 km and beyond. At long distances, GLA achieves higher measured speedup partly because the exact baseline traversals are costlier on the denser network; the deterministic schedule remains conservative for GLA (Figure 5).

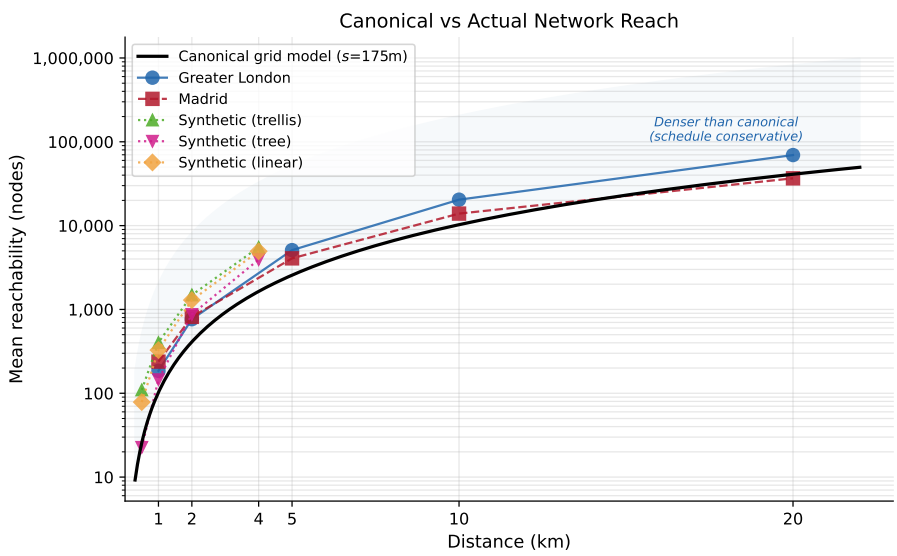

Figure 5: **Canonical vs. actual network reach.** Mean reach (log scale) vs. analysis distance for Greater London, Greater Madrid, and three synthetic topologies, compared against the canonical grid model ($s = 175$ m, black line). Greater London lies consistently above the canonical curve (the schedule over-samples). Greater Madrid falls below at longer distances, reflecting sparser peripheral areas. Both validation networks achieve $\rho \geq 0.95$ at all distances (Tables 3–4).

Table 5: Deterministic sampling schedule by analysis distance ($\varepsilon = 0.06$, $\delta = 0.1$, canonical grid spacing $s = 175\,\mathrm{m}$). Both closeness and betweenness use the same schedule. Sampling is exact ($p = 100\%$) for distances up to 4 km. Speedup shown is theoretical ($1/p$); measured wall-clock speedup is lower at near-unity p due to sampling overhead and emerges at 10 km and beyond.

Distance	**Canonical reach**	k	p (%)	**Speedup** ($1/p$)
1 km	103	1,059	100	1×
2 km	410	1,252	100	1×
5 km	2,565	1,506	58.7	1.7×
10 km	10,258	1,699	16.6	6.0×
20 km	41,033	1,891	4.6	21.7×

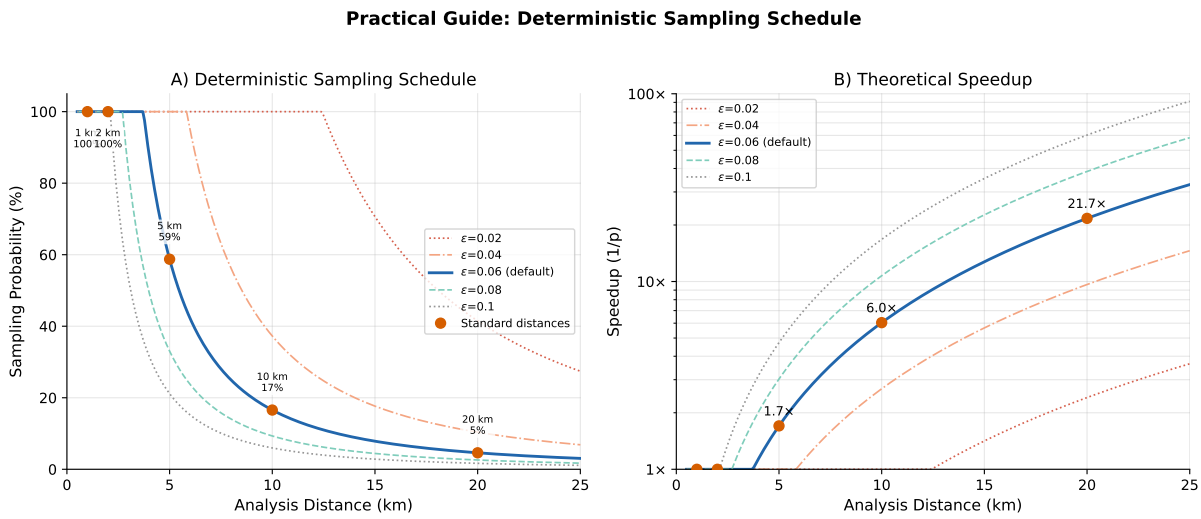

Figure 6: **Practical guidance.** (A) Deterministic sampling probability p vs. analysis distance under the canonical grid model ($\varepsilon = 0.06$, $\delta = 0.1$), with standard analysis distances annotated. (B) Theoretical speedup $1/p$ vs. distance; measured wall-clock speedup is lower at near-unity p due to sampling overhead (see Figure 4).

Because $k \propto 1/\varepsilon^2$, halving ε quadruples the required sources. The default $\varepsilon = 0.06$ is recommended for publication-quality results; $\varepsilon = 0.1$ suffices for exploratory analysis ($\rho > 0.99$ on synthetic networks; Figure 1). The schedule is calibrated to betweenness, so users computing only closeness may use a larger ε.

Sampling is not appropriate when exact centrality values are required or when the network is small ($< 1{,}000$ nodes). For untested network morphologies, a pilot comparison at one representative long distance (e.g., 10 km) with and without sampling can verify the schedule: $\rho > 0.95$ confirms adequate performance; if not, reducing ε restores accuracy.

Limitations. First, we assume uniform random sampling of source nodes. Non-uniform strategies (e.g., stratification or importance weighting) could achieve higher accuracy with fewer samples, but require proxies for node importance that are unavailable before computation.

Second, the deterministic schedule substitutes canonical grid reach $r = \pi d^2/s^2$ for actual network reach. This makes the schedule network-agnostic: denser networks are over-sampled, while sparser networks rely on the Hoeffding slack for accuracy (confirmed by Madrid; Figure 5). Networks with inter-node spacing substantially exceeding 175 m (e.g., rural road networks) would warrant separate validation.

Third, the model treats the network as statistically homogeneous: accuracy depends on distance rather than local structure. Validation on the GLA and Madrid networks, which include dense urban cores and sparse suburbs, suggests the schedule is conservative overall, but accuracy may vary locally.

Further scope limitations: results are averaged over three random seeds (standard deviations are small); only shortest-path distance is tested (angular distance warrants separate validation); validation covers two European cities (other morphologies would strengthen external validity); and betweenness shows thinner margins than closeness, with minimum $\rho = 0.9767$ (Madrid, 20 km).

Future work. Adaptive per-node sampling (varying p with local reach) could improve efficiency in heterogeneous networks. Online stopping criteria (Borassi and Natale, 2019) could replace a priori calibration but require new theory for localised centrality. Variance-dependent bounds (Pellegrina and Vandin, 2023) offer a path to tighter schedules on well-structured networks. Validation for angular distance and other centrality measures would broaden applicability.

Anchoring a source-sampling schedule to a canonical grid model trades maximum speedup for robustness and cross-network comparability. The resulting error structure is well suited to urban analysis: high-centrality nodes receive the most precise estimates, while low-reach peripheral nodes approach exact computation. Across the Greater London and Greater Madrid networks at 1–20 km, the schedule achieves $\rho \geq 0.97$ with speedups of up to 8.0×, making multi-scale centrality analysis practical at metropolitan scales.

The model is implemented in the open-source `cityseer` Python library.[2] Analysis scripts and validation data are available in the repository's `analysis/sampling/` directory.

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A Technical Details

Synthetic network generation. Synthetic networks are generated using tile-based templates. Each template defines a connectivity pattern within a single tile; tiles are then arrayed to create networks of the desired extent.

Trellis A regular grid with diagonal connections, representing highly connected urban cores. Average degree ≈ 4.

Tree A hierarchical branching structure where each node connects to at most 3 others. Represents suburban cul-de-sac development. Average degree ≈ 2.3.

Linear A constrained grid with fewer cross-connections than trellis. Intermediate connectivity. Average degree ≈ 3.

Networks are tiled 24×24 to achieve approximately 12 km extent, then the largest connected component is extracted. Node positions are perturbed slightly to avoid perfectly regular spacing.

GLA network preprocessing. The Greater London Authority road network is sourced from Ordnance Survey Open Roads data (GeoPackage format).

Preprocessing steps:

1. Remove invalid and empty geometries
2. Explode multi-part geometries
3. Build NetworkX graph from edges
4. Remove filler nodes (degree-2 nodes that do not represent junctions)
5. Remove dangling nodes (dead-end segments < 20 m)
6. Consolidate nearby nodes (threshold 1 m)
7. Extract largest connected component

Final network: $\approx$ 156,285 nodes.

Correctness verification against NetworkX. We verified the Rust-based `cityseer` implementation against NetworkX (Hagberg et al., 2008) across all three synthetic topologies, distance thresholds from 200 m to 3200 m, both distance heuristics, and both metrics. The maximum per-node difference

$$\Delta_v = |C_{\text{cityseer}}(v) - C_{\text{NetworkX}}(v)| \tag{10}$$

was $\max_v \Delta_v < 10^{-6}$ in all configurations, attributable to floating-point precision.

Horvitz–Thompson estimator. When estimating population totals from samples, two classical approaches are the Horvitz–Thompson (H–T) estimator and the Hájek ratio estimator. Under Bernoulli (independent) source sampling with inclusion probabilities π_i, the Horvitz–Thompson estimator (Horvitz and Thompson, 1952) estimates a population total as $\hat{T}_{HT} = \sum_{i \in S} y_i/\pi_i$, where S is the set of sampled sources. This inverse-probability weighting yields an unbiased estimator whenever $\pi_i > 0$ for all sampled units.

The Hájek estimator (Hájek, 1971) instead uses a ratio form,

$$\hat{T}_H = N \cdot \frac{\sum_{i \in S} y_i/\pi_i}{\sum_{i \in S} 1/\pi_i}, \tag{11}$$

which is often lower-variance but can be slightly biased in finite samples (Cochran, 1977; Särndal et al., 2003). In `cityseer`, we use Horvitz–Thompson-style inverse-probability weighting because it matches the paper's $1/p$ rescaling description and extends directly to unequal inclusion probabilities (e.g., when combining a base sampling probability with per-source importance weights).

Implementation in `cityseer`. The sampling model is implemented in the `cityseer` Python library (version 4.23+). Sampling is enabled via the `sample=True` flag:

```
from cityseer.metrics import networks

gdf = networks.closeness_shortest(
    network_structure,
    nodes_gdf,
    distances=[500,1000,2000,5000,10000],
    sample=True
)

gdf = networks.betweenness_shortest(
    network_structure,
    nodes_gdf,
    distances=[500,1000,2000,5000,10000],
    sample=True
)
```

The default error tolerance ($\varepsilon = 0.06$) and confidence ($\delta = 0.1$) are set in `cityseer.config`. Helper functions for custom workflows are available in `cityseer.config`, including:

- `compute_hoeffding_p`: Hoeffding bound from mean reach
- `compute_distance_p`: deterministic p from distance via canonical grid model

Reproducibility. All analysis scripts and validation results are available at analysis/sampling/. The current script pipeline runs sequentially from `00_generate_cache.py` through `05_generate_macros.py` via `run_all.py`, with each script's outputs documented in its header.